

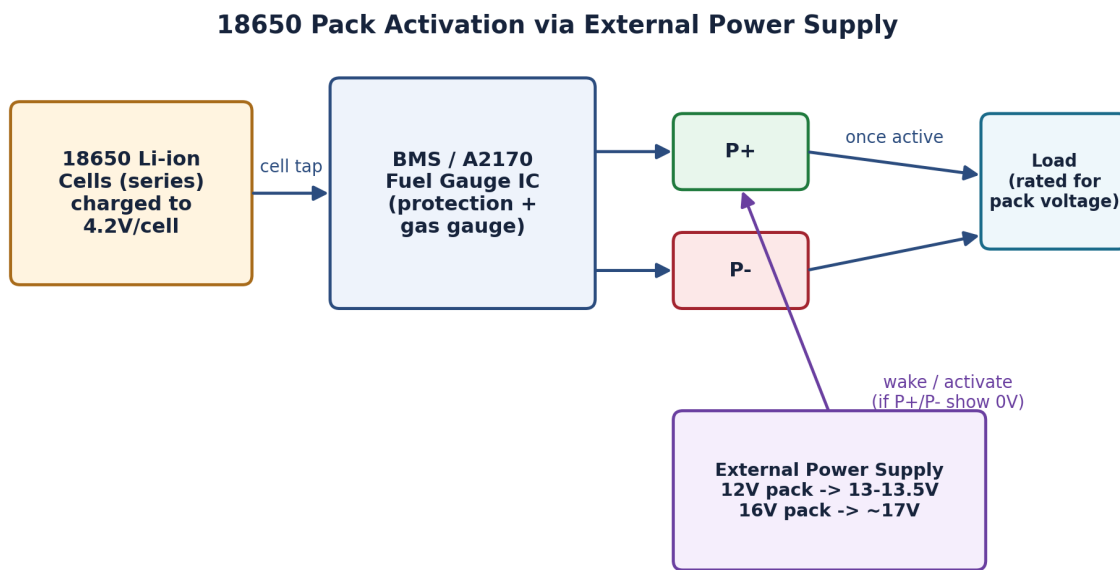
A2170 BMS Fuel Gauge (Gas Gauge) IC

Where This A2170 Fuel Gauge Was Found

Mostly, this BMS IC is found in local laptop batteries — for example, Lapcare and Apex. I found it in these two.

How to Turn On a Full Battery

First, charge each 18650 lithium-ion cell fully to 4.2 V. Then check the voltage at P+ (positive) and P- (negative). If no voltage appears, apply power from an external supply: for a 12 V battery pack, use 13–13.5 V; for a 16 V pack, add one extra volt, meaning about 17 V externally. The battery becomes active once the BMS has no fault. Once the pack is active, connect any load to P+ and P- with a voltage rating that matches your pack, since the series voltage across the cells may not equal the voltage at P+ and P- due to a voltage drop in the BMS or a fault. Check first that everything is working fine before moving on to the communication part.



Check P+/P- with a multimeter first. If 0V, BMS may be latched off due to a fault - apply the external supply briefly to wake it, then re-check before connecting any load.

Fig. 1 — Waking a latched-off pack with an external supply before connecting a load.

How to Communicate With the BMS Using SMBus

First, fully charge each 18650 lithium-ion cell to 4.2 V. Then check the voltage at P+ (positive) and P- (negative). If no voltage appears, apply power from an external supply: for a 12 V battery pack, use 13–13.5 V; for a 16 V battery pack, use about 17 V externally. The battery becomes active when the BMS has no fault. Once the pack is active, connect a load to P+ and P- with a voltage rating that matches the pack, because the series voltage across the cells may not match the voltage at P+ and P- due to a voltage

drop in the BMS or a fault.

After confirming that everything is working properly, move on to communication with the BMS over SMBus. In most cases, you need three pins: SDA, SCL, and GND, connected to the MCU's SDA, SCL, and GND pins. In some cases, you may also need signal pull-ups, so add 4.7 k Ω resistors to 3.3 V on both SDA and SCL. First, upload code that detects an address, such as 0x0B. Then proceed by sending commands to the register bank to fetch the data you need.

SMBus Wiring: MCU <-> BMS / A2170 Fuel Gauge (addr 0x0B)

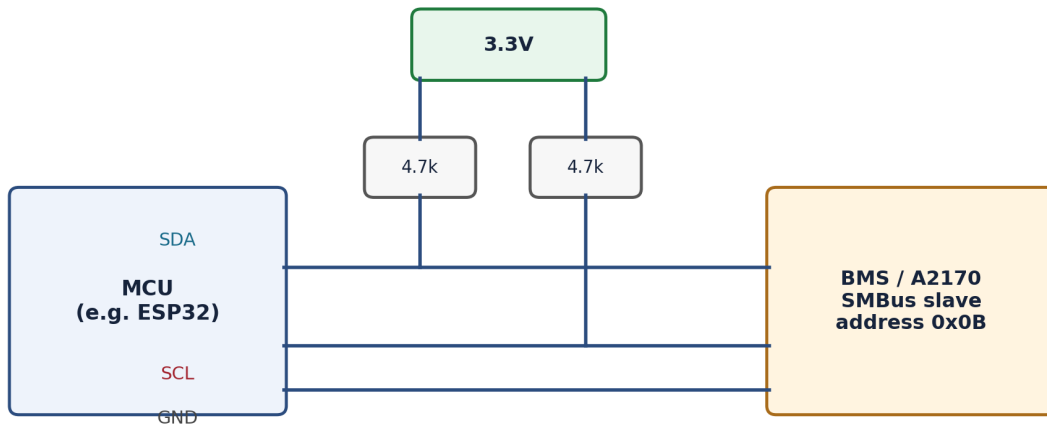


Fig. 2 — SMBus connections between the MCU and the BMS, with 4.7k Ω pull-ups to 3.3V.

SMBus-capable battery management ICs (including the A2170 and the Texas Instruments BQ-series ICs it emulates) support commands over SMBus to control the battery pack's output state.